TASK 1 | MAIN DISTRIBUTION, PROTECTION AND METERING

REV04 | 380/220 V, 3-phase 4-wire + PE | Single-line diagram, not to scale | Normal supply only

TX-A | CLP 1500 kVA

Utility HV protection
11 kV / 380 V
uk = 6% assumed

ACB-A | 4P draw-out
MTZ2 25H1 / 6.0X LSIG
2500AF; Ir 2250 A; 66 kA

4-pole sensors > LSIG > trip
Isd 4500 A; tsd 0 s; I2t OFF
Ig 500 A; tg 0.2 s, I2t OFF

MSB-A | 2500 A section

BC | 4P, N.O.
2500AF / 2250AT
MTZ2 25H1, 66 kA

A-FL | 1600AF
Ir 1440 A; 66 kA
KTC2500 busway

A-PL | 800/720 A
4P, 70 kA
Plant / public

F1 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F2 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F3 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F4 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F5 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

TPN MCCB BEFORE each tenant meter:
O1/O2 100 A; O3 125 A + CT meter.
F1/O1 63 A; all tenant MCCBs70 kA.

TX-B | CLP 1500 kVA

Utility HV protection
11 kV / 380 V
uk = 6% assumed

ACB-B | 4P draw-out
MTZ2 25H1 / 6.0X LSIG
2500AF; Ir 2250 A; 66 kA

4-pole sensors > LSIG > trip
Isd 4500 A; tsd 0 s; I2t OFF
Ig 500 A; tg 0.2 s, I2t OFF

MSB-B | 2500 A section

B-FL | 1600AF
Ir 1440 A; 66 kA
KTC2500 busway

B-PL | 800/720 A
4P, 70 kA
Plant / public

F6 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F7 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F8 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F9 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

F10 — 400AF / Ir 320 A, 4P, 70 kA — M O1 — M O2 — O3 — M CT

TPN MCCB BEFORE each tenant meter:
O1/O2 100 A; O3 125 A + CT meter.
F1/O1 63 A; all tenant MCCBs70 kA.

F1 / OFFICE 1 | DETAILED FEEDER

From F1 320 A floor MCCB
MS-O1: ComPacT NSX100H
C10H4TM063, 4P4D TM63D
Ir 63 A; Icu/Ics 70 kA
Suitable for isolation

M CLP tenant meter
Immediately after MS-O1

4C35 mm2 PVC Cu
+ separate CPC16 mm2
L = 31.5119 m
28.5119 plan + 3 m allowance

DB-O1: 100 A 4P isolator
15 x 1P+N Type A 30 mA RCBO
7 lighting + 8 power circuits

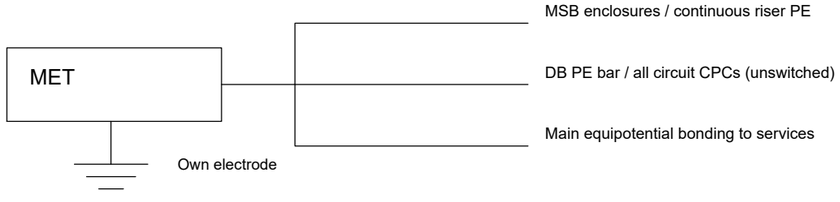
PE continuous; neutral switched with phases.
No customer N-PE link at DB.

ACB-A / ACB-B / BC INTERLOCK

Electrical AND mechanical interlock
Normal: CLOSED / CLOSED / OPEN
Transfer from A: CLOSED / OPEN / CLOSED
Transfer from B: OPEN / CLOSED / CLOSED
ALL THREE CLOSED: PROHIBITED

Open/isolate failed incomer before closing BC.
Before transfer, shed load to <=1480.9 kVA
(2250 A at 380 V); no full-load N-1.
Utility return: open BC before restoring incomer.

CUSTOMER EARTHING



BASIS AND FAULT DUTY

CLP GuideToSupply v10: sections 2, 3, 7.4, 11.
MSB PFC <=56.92 kA incl. transfer motors.
Own MET/electrode; any CLP bond only by agreement, >=70 mm2 Cu. See Task 3 for loop, disconnection and conductor checks.
CT meters: CLP ratios +16 A HRC voltage fuses; Guide sections9.11 and9.16.

2672.216 kVA building demand | 1336.108 kVA / 2030 A per normal section | CT/sensor arrangement subject to CLP acceptance